

CPR Esoft Quiz 1 – Answer Key and Rationale

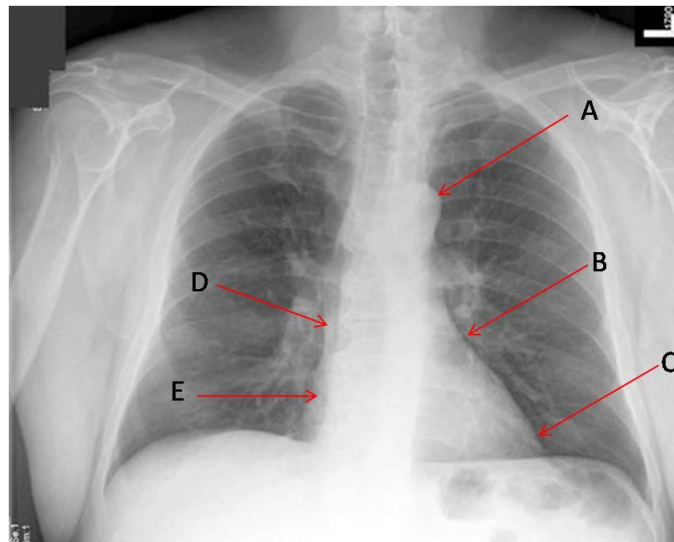
Question #: 1

A 50-year-old man is brought to the emergency department by the paramedics because of difficulty breathing immediately after being involved in a motor vehicle collision. His blood pressure is 70/50 mmHg, pulse is 105/min and respirations are 22/min. Venous return to the heart is blocked. In the attached radiograph shown, which of the following areas is most likely affected?

- A. A
- B. B
- C. C
- ✓D. D
- E. E

Rationale: Venous return to the heart is through the inferior vena cava and superior vena cava. In the radiograph, D indicates the area of the superior vena cava.

Attachment:



chest x ray.jpg

Question #: 2

A newborn boy is examined by a physician. Auscultation of his chest shows a holosystolic heart murmur. Ultrasonography shows an open structure connecting the aorta and the pulmonary trunk. Which of the following is the most likely embryological origin of this structure?

- A. Left dorsal aorta
- B. Right dorsal aorta
- C. Left fifth aortic arch
- ✓D. Left sixth aortic arch
- E. Right sixth aortic arch

Rationale: The ductus arteriosus connects the pulmonary trunk to the aorta. This is derived from the left sixth aortic arch.

Question #: 3

A premature newborn boy is admitted to the neonatal intensive unit because of severe respiratory distress. Despite medical intervention, He dies within 24 hours. The postmortem examination shows the cause of death is respiratory distress syndrome. There is an incidental finding of an atrial septal defect (ASD) that has little to no hemodynamic/clinical significance. Which of the following types of ASD is most likely?

- A. Ostium secundum
- ✓B. Probe patent oval foramen
- C. Common atrium
- D. Endocardial cushion with ostium primum
- E. Sinus venosus

Rationale: A probe patent foramen ovale is one where there is a very small defect in the atrial septum to failure of complete closure of the foramen ovale. This defect is hemodynamically stable and of no clinical significance.

Question #: 4

A 59-year-old woman comes to the physician for a routine examination. Physical examination shows a systolic murmur which is loudest at the second intercostal space on the right. Echocardiography shows valvular stenosis. Which of the following chambers or vessels most likely receives the blood that is being ejected through the narrowed opening?

- A. Left ventricle
- B. Right ventricle
- ✓C. Aorta
- D. Pulmonary trunk
- E. Left atrium

Rationale: The valve that is auscultated in the right second intercostal space is the aortic valve. This is the valve between the left ventricle and the ascending aorta.

Question #: 5

A 75-year-old woman is brought to the emergency department due to a feeling of pressure-like chest pain beneath the sternum, which radiates to her left arm. An ECG indicates a myocardial infarction, along with a right bundle branch block. Physical examination shows a pronounced second heart sound. Which pair of heart valves' closure is primarily responsible for generating this second heart sound?

- ✓A. Aortic and pulmonary
 - B. Aortic and tricuspid
 - C. Tricuspid and mitral
 - D. Mitral and pulmonary
 - E. Tricuspid and pulmonary
-

Question #: 6

A 14-year-old boy is brought to the physician for a routine examination. Physical examination shows a regurgitation murmur of his bicuspid valve. Into which of the following chambers or vessels is the blood most likely regurgitating?

- ✓A. Left atrium
- B. Right atrium
- C. Pulmonary trunk
- D. Aorta
- E. Left ventricle

Rationale: The bicuspid valve lies between the left atrium and the left ventricle. If there is an incompetent bicuspid valve, blood will regurgitate back into the left atrium.

Question #: 7

A 48-year-old man comes to the physician because of a 4-month history of shortness of breath on exertion. Coronary angiography shows severe occlusion of the anterior interventricular artery. He is taken for coronary bypass surgery. Which of the following cardiac veins most likely

accompanies the blocked artery?

- A. Middle
- B. Anterior
- C. Posterior
- D. Coronary sinus
- ✓E. Great

Rationale: The great cardiac vein runs alongside the anterior interventricular artery, while the middle cardiac vein runs alongside the posterior interventricular artery.

Question #: 8

A 56-year-old man is admitted to the hospital for surgery to correct a congenital atrial septal defect. During the surgery, the surgeon must be careful to stay away from the blood supply to the conducting system of the heart including the supply to the SA and AV nodes. Which of the following vessels should most likely be avoided?

- A. Anterior interventricular
- B. Left coronary
- ✓C. Right coronary
- D. Circumflex
- E. Posterior interventricular

Rationale: The right coronary artery gives off the sinu-atrial nodal branch which passes posteriorly around the superior vena cava to supply the sinu-atrial node. As it continues the diaphragmatic surface of the heart, it gives a small branch to the AV node.

Question #: 9

A 50-year-old man is brought to the emergency department by his wife because of a 3-hour history of substernal chest pain that radiates into his neck and left upper limb. ECG shows ST elevation in the precordial leads. A diagnosis of myocardial infarction is made. Which of the following structures most likely carries the visceral afferent fibers responsible for the patient's pain?

- A. Intercostobrachial nerve
- ✓B. Cardiopulmonary nerves
- C. Greater splanchnic nerves

- D. Phrenic nerve
- E. Grey rami communicans

Rationale: The cardiopulmonary nerves supply postganglionic sympathetic to the cardiac plexus to supply the heart. Visceral afferent fibers from the heart travel back to the spinal cord with these sympathetic fibers.

Question #: 10

A 2-day-old girl is admitted to the cardiology unit because of a 6-hour history of labored breathing and cyanosis. Echocardiography was performed, which shows a congenital heart defect that results in a right-to-left shunt. Which of the following conditions is most likely to produce this type of shunt at birth?

- ✓A. Tetralogy of Fallot
- B. Atrial septal defect
- C. Ventricular septal defect
- D. Patent ductus arteriosus
- E. Patent foramen ovale

Rationale: A cyanotic congenital heart defect is due to a right to left shunt where deoxygenated blood is shunted into the systemic circulation. This leads to decreased amount of oxygenated blood in the systemic circulation and cyanosis. Tetralogy of Fallot is one such cyanotic heart condition.

Question #: 11

A 50-year-old man is brought to the emergency department because of a 3-hour history of substernal chest pain that radiates into his neck and left upper limb. ECG shows ST elevation in the precordial leads. A diagnosis of myocardial infarction is made. Which of the following structures is most likely responsible for the pain being referred to the upper limb?

- ✓A. Intercostobrachial nerve
- B. Cardiopulmonary nerves
- C. Greater splanchnic nerves
- D. Phrenic nerve
- E. Grey rami communicans

Rationale: The intercostobrachial nerve is the nerve through which pain from a myocardial infarction is referred to the upper limb.

Question #: 12

A 2-year-old boy is brought to his physician by his mother because of a cold. His mother says that he has had frequent episodes of “the cold” since birth. Physical examination show decreased breath sounds in both lungs and heart sounds heard best on the right chest. Chest x-ray show the apex at the right 4th intercostal space. Defect in which of the following processes resulted his findings?

- A. Lateral folding
- B. Head to tail folding
- C. Formation of pleuropericardial folds
- ✓D. Dextro-looping
- E. Canalization of primordial heart tube

Rationale: Correct answer. Dextro-looping is the correct. This takes place when the heart rotates to the right, instead off the left. This condition is usually seen when there is a mutation on the chromosome 12. The disc-like embryo then undergoes a process of folding, in which both the cranial and lateral parts of the embryo fold ventrally. This brings the heart-forming region to a ventral position.

Incorrect answers. Lateral folding and Head to tail folding are incorrect.

Formation of pleuropericardial folds is incorrect. Pleuropericardial folds- flaps of body wall at the level of the primitive pericardial cavity which grow medially and fuse with the ventral surface of the foregut mesoderm. This subdivides the primitive pericardial cavity into the definitive pericardial and two pleural cavities.

Canalization of primordial heart tube is incorrect. Heart tube undergoes canalization during development.

Question #: 13

A 58-year-old man comes to the physician because of a 2-month history of difficulty swallowing. Imaging studies show an enlargement of the most posteriorly located chamber of heart which is impinging on the esophagus resulting in his symptoms. Which of the following parts of the heart is most likely in a position to cause this impingement?

- ✓A. Left atrium
- B. Right atrium
- C. Left ventricle
- D. Right ventricle
- E. Aortic arch

Rationale: The left atrium is posteriorly located. It lies anterior to the esophagus; therefore, enlargement can impinge on the esophagus, leading to difficulty swallowing.

Question #: 14

A 55-year-old man comes to the physician for a follow-up examination. He performs poorly on an exercise stress test. Coronary angiography shows significant blockage of one of the major arteries supplying his heart. He is sent for a coronary artery bypass procedure. During the surgery, after the chest is opened using a midline sternotomy (splitting of the sternum), the pericardial sac is opened by a longitudinal incision. If a horizontal incision has been used on the pericardium, which of the following structures would be most likely be at risk?

- A. Vagus nerve
- B. Internal thoracic artery
- C. Azygos vein
- D. Thoracic duct
- ✓E. Phrenic nerve

Rationale: The phrenic nerves descend on the left and right sides of the fibrous pericardium. They pass through and innervate the fibrous pericardium on their way to innervating the diaphragm.

Question #: 15

A 78-year-old female presented with edema of the left upper limb due to poor venous return. Examination revealed an aneurysm of the ascending aorta that was impinging on a large vein lying immediately antero-superior to it. Which of the following veins is most likely being compressed?

- A. Azygos
- B. Internal thoracic
- ✓C. Left brachiocephalic
- D. Left superior intercostal
- E. Right brachiocephalic

Rationale: Correct answer. Left brachiocephalic is the correct answer. Left brachiocephalic vein has a horizontal course and is related anteriorly and superior to aorta. This vein could be compressed in the case of aortic aneurysm.

Incorrect answers. Azygos vein is incorrect. The azygos vein is a vein running up the right side

of the thoracic vertebral column draining itself towards the superior vena cava. Internal thoracic vein is incorrect. Internal thoracic vein is a vessel that drains the chest wall and breasts. Bilaterally, it arises from the superior epigastric vein, accompanies the internal thoracic artery along its course, and terminates in the brachiocephalic vein. Left superior intercostal vein is incorrect. The left superior intercostal vein drains the 2nd and 3rd posterior intercostal veins on the left side of the body. It drains into the azygos vein or left brachiocephalic vein. It may also communicate with the accessory hemiazygos vein. Right brachiocephalic vein is incorrect. It gets the venous blood from the right subclavian and right internal jugular vein and has a vertical course and drains into the superior vena cava.

Question #: 16

A 56-year-old man is brought to the emergency department by his son because of a 2-day history of difficulty breathing. Two months ago, he was diagnosed with a mediastinal tumor. Imaging studies show that cancer has invaded the nerve that travels anterior to the hilum of the lung. Which of the following nerves is most likely affected?

- A. Vagus
- B. Cardiopulmonary
- ✓C. Phrenic
- D. Greater thoracic splanchnic
- E. Intercostobrachial

Rationale: The phrenic nerve arises from the C3-C5 spinal cord segments. It descends through the thorax to supply the pericardium and the diaphragm. As it descends, it passes anterior to the hilum of the lung.

Question #: 17

A 2-month-old infant is brought to the physician due to poor feeding and rapid breathing. Physical examination shows a systolic murmur and cyanosis. Echocardiography shows a large ventricular septal defect (VSD) and an overriding aorta. Given these findings, which of the following embryological events is most likely responsible for this infant's condition?

- A. Abnormal development of endocardial cushions
- ✓B. Incorrect division of the truncus arteriosus
- C. Failure of the foramen ovale to close after birth
- D. Abnormal development of the sinus venosus
- E. Insufficient formation of the bulbus cordis

Question #: 18

A 45-year-old woman comes to the physician because of a 2-week history of worsening left sided upper limb edema. She has a history of left breast cancer for which she received surgical treatment including left axilla lymph node dissection. Physical examination show 1+ pitting edema in the left upper extremity with no erythema or increased warmth. Which of the following vessels is most likely affected in this case?

- ✓A. Lymphatic capillaries
- B. Fenestrated capillaries
- C. Sinusoidal capillaries
- D. Elastic artery
- E. Large vein

Rationale: Lymphatic capillaries are specialized microvessels that collect excess interstitial fluid, proteins, and waste products from tissues. They are characterized by their closed-ended structure and permeable walls, allowing fluid to enter but not exit. This collected fluid, now termed lymph, is transported through larger lymphatic vessels and eventually returned to the bloodstream via the subclavian veins.

Cleveland Clinic

In cases of lymphedema, such as this patient's, the compromised lymphatic capillaries cannot effectively drain the accumulated lymph, leading to fluid retention and the characteristic swelling observed in the affected limbs.

Question #: 19

A 64-year-old woman is brought to the emergency department because of a sudden onset of chest pain, nausea, vomiting, and blurred vision. She has a history of essential hypertension and is non-adherent to her medication. Her blood pressure is 180/120 mm Hg, and pulse is 92/min. Physical examination shows crackling sounds in both lung fields. She is diagnosed with hypertensive emergency and started on hydralazine (an antihypertensive medication). This medication is known to regulate vascular tone by causing widening of blood vessels (vasodilation) by acting directly on the smooth muscle of the vessel wall. Which of the following best identifies the layer of the vessel wall in which this muscle type predominates?

- ✓A. Media
- B. Epicardium

- C. Adventitia
- D. Intima
- E. Endocardium

Rationale: Muscle and elastic fibers can be found in all 3 layers of the vessel walls, but the tunica media has the greatest preponderance of muscle fibers. Epicardium and endocardium are layers of the heart wall, not the vessel wall.

Question #: 20

A 78-year-old woman comes to the physician because of a 2-week history of brief fainting episodes while climbing steps and occasional lightheadedness during prolonged walking, and a 3-month history of worsening shortness of breath during regular house cleaning activities. She was diagnosed with angina pectoris (chest pain brought on by exertion and relieved with rest) 3 months ago. Physical examination shows bilateral swollen ankles and an irregular heartbeat. Cardiac imaging studies show a "floppy" aortic valve. Examination of a biopsy specimen obtained from the affected valve show pathological accumulations of proteoglycans with degradation of collagen and elastic fibers. Which of the following best characterizes the layer of the heart valve in which the accumulated extracellular matrix component is normally found?

- A. Simple projections of the tunica intima
- B. Lined by endothelium
- C. Lacks collagen fibers
- D. Lacks elastic fibers
- ✓E. Functions as a shock absorber

Rationale: The aortic valve is a semilunar valve found in the heart, and separating the left ventricle from the aorta. The valves of the heart are made up of 3 layers, the fibrosa, middle spongiosa and atrialis or ventricularis layer. Both the atrialis/ventricularis and fibrosa layers contain elastic and collagen fibers (types I and III) and are lined by endothelium. Although both collagen and elastic fibers are found in the spongiosa (middle layer), this layer additionally contain an abundance of ground substance in the form of GAGS and PGs. It is the only layer that contains GAGS and PGS, which allows it to act as a shock absorber. Loss of collagen fibers will result in a floppy valve.

Venous valves are simple projections of the tunica intima. This question is about heart valves, not venous valves.

Question #: 21

A 42-year-old man is brought to the emergency department because of a 1-hour history of

seizures, and a 2-day history of headaches and disorientation. He has a history of a viral illness 2 weeks ago. His temperature is 104°F (40°C), and pulse is 102/min. Physical examination shows an unconscious, febrile man. He dies despite appropriate medical care. Histopathological examination of biopsy specimens of the brain shows necrotic changes to the endothelial cells that line the lumen of the vessel that forms part of the blood-brain barrier. Which of the following best identifies an alternative location for the most likely affected type of blood vessel?

- A. Ciliary processes of the eye
- B. Glomeruli of the kidneys
- C. Spleen
- D. Liver
- ✓E. Muscle

Rationale: Type I capillaries forms barriers in the body. Here, the type I capillary is affected. Another location is muscle tissue.

- A - type II capillary
- B - type II capillary
- C - type III capillary
- D - type III capillary

Question #: 22

A 62-year-old man is brought to the emergency department because of a 2-day history of fevers, chills, and generalized weakness. Physical examination shows a weak man that is warm to the touch. Laboratory studies show an elevation in both the white blood cell count, and biomarkers for endothelial activation such as von Willebrand factor. Which of the following is most likely a normal function of the activated tissue?

- A. Production of platelets for blood clotting
- ✓B. Secretion of vasoactive substances such as nitric oxide
- C. Production of leukocytes
- D. Production and secretion of erythrocytes
- E. Maintains a pro-thrombogenic surface at all times

Rationale: In this instance, the endothelial cells are activated, which is commonly seen in severe infections. Endothelial activation is a shift in the routine function of endothelial cells to that of more a pro-inflammatory, pro-thrombogenic function. Normally, endothelial cells produce nitric oxide which promotes vasodilation. Platelets, leukocytes, and erythrocytes are produced in the bone marrow. Endothelial cells promote an anti-thrombogenic surface, not a pro-thrombogenic surface. They only become pro-thrombogenic when activated, under stressful situations.

Question #: 23

A 52-year-old man comes to the physician for a follow up examination. He has a 15-year history of high blood pressure. His pulse is 82/min, blood pressure is 142/98 mm Hg, and respirations are 12/min. Physical examination shows a split S2 sound. Which of the following cardiac events is most likely associated with this heart sound during a cardiac cycle?

- A. Closure of the mitral valve
- B. Opening of the tricuspid valve
- ✓C. Closure of the aortic and pulmonic valves
- D. Inrushing blood into the ventricle during atrial systole
- E. Rapid ejection of blood into the aorta

Rationale: The heart sounds are due to vibration of blood with valve closure; however, it is easier just to associate the sounds with valve closures itself. (Normally healthy valves do not make sounds). The first heart sounds are associated with the AV valves closure and arterial valves (aortic, and pulmonic) closure are associated with the second heart sounds.

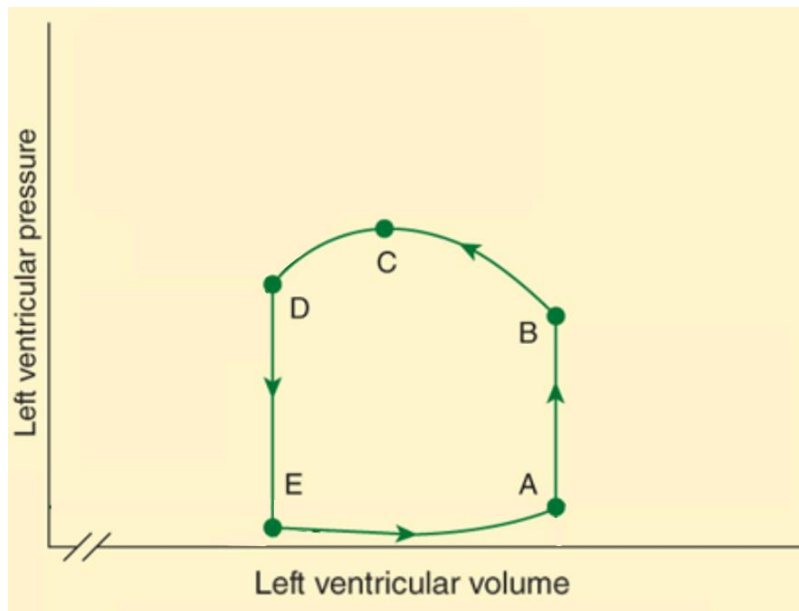
Question #: 24

A 35-year-old man comes to the physician because of a 2-month history of shortness of breath. He has had rheumatic fever in his childhood. His vital signs are within normal limits. Physical examination shows an opening snap followed by a rumbling mid-diastolic murmur in the mitral area. Echocardiography shows a stenotic mitral valve. Which of the following labelled points best represents the opening of the affected valve in the attached pressure-volume loop figure?

- A. A
- B. B
- C. C
- D. D
- ✓E. E

Rationale: The mitral or bicuspid valve is located between the left atrium and the left ventricle. In normal conditions blood flows through the open mitral valve during diastole due to contraction of the left atrium, and it closes during systole on contraction of the left ventricle. The valve opens when there is greater pressure in the left atrium than ventricle, which corresponds to point E on the graph (end of systole and onset of diastole).

Attachment:



Capture 2.PNG

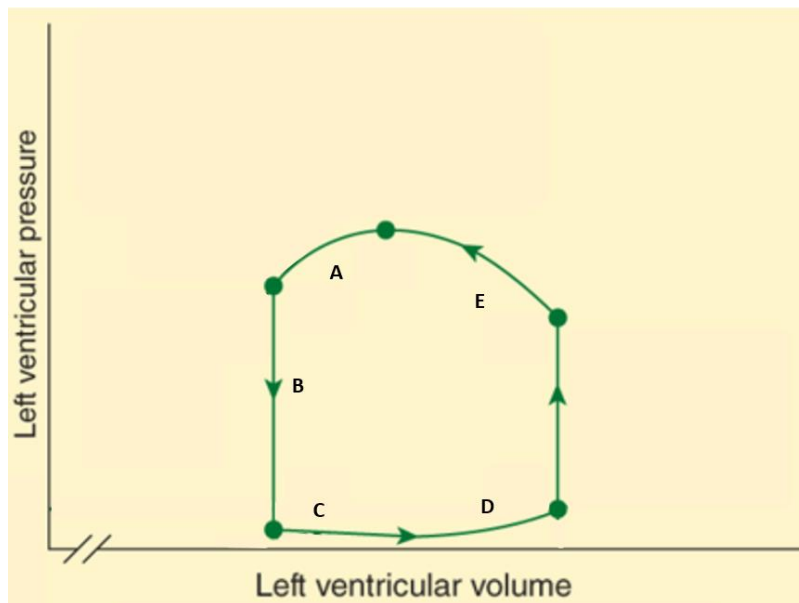
Question #: 25

A 65-year-old woman is brought to the emergency department because of a 2-hour history of chest pain and shortness of breath. She has a 15-year history of hypertension and type 2 diabetes mellitus. The pain started when the patient was walking up the hill. It was in the anterior chest wall with radiation to the left upper limb and the left jaw. Her pulse is 86/min, blood pressure is 148/94 mmHg, and respirations are 18/min. Physical examination shows crepitations in the basal lungs and a loud S3 in the mitral area. The pressure-volume loop of the patient is shown in the attached figure. Which of the labeled parts of the graph is most likely related to the abnormal heart sound heard in this patient?

- A. A
- B. B
- ✓C. C
- D. D
- E. E

Rationale: C and D in the graph represent the period of ventricular filling and atrial contraction. The S3 sound is related to the rapid ventricular filling phase particularly, which is represented during the period labeled with letter C, just after the opening of the mitral valve.

Attachment:



Picture3.png

Question #: 26

A 62-year-old man is brought to the intensive care unit because of overwhelming hypotension and fever. The patient has received 4 L of normal saline bolus fluids. Empirical antibiotics were begun with no improvement in his condition. His blood pressure is 60/30 mm Hg, pulse is 112/min, temperature is 40.6° C (105° F), and respiratory rate is 23/min. The physician orders intravenous norepinephrine. Which of the following is a direct therapeutic effect of norepinephrine in this clinical scenario?

- A. Bradycardia
- B. Bronchoconstriction
- ✓C. Coronary vasodilation
- D. Decreased inotropy
- E. Peripheral vasodilation

Rationale: Dilation of the coronary resistance vessels because of increased myocardial metabolism after adrenaline. that the primary effect of adrenaline may be a coronary vaso-constriction which is followed by a vasodilatation as the metabolic effects of adrenergic stimulation become apparent.

Question #: 27

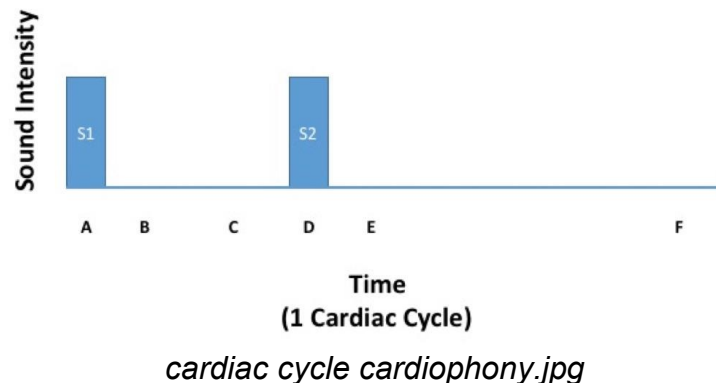
A 59-year-old man comes to the physician because of 3-week history of breathlessness and severe tiredness on exertion. His pulse is 90/min and blood pressure is 130/50 mm Hg. Physical

examination shows a murmur on auscultation that is made worse during maneuvers to increase peripheral resistance, such as squatting and hand grip exercises. Echocardiography shows an increased left ventricular end diastolic volume. Using the attached diagram of a cardiac cycle cardiophony, during which of the following phases is the murmur most likely to be heard loudest?

- A. A
- B. B
- C. C
- D. D
- ✓E. E
- F. F

Rationale: Let us look at the symptoms step by step. Tired on exertion suggest that the tissues are not receiving enough blood. Palpitations are associated with many diseases, so it is not too helpful here. The blood pressure gives it away, with a wide pulse pressure and low diastolic. A murmur narrows things down to heart valve dysfunction. Exacerbation of the murmur with systemic resistance suggest aortic etiology and not atrial. EDV (preload) is increased in aortic and mitral regurgitation, and not with stenoses. So, on re-evaluation- we select aortic regurgitation based on the blood pressure (pulse pressure), EDV, and suggestion of aortic problems. Now using the mnemonic MRARD we know that aortic regurgitation involve diastole, the sound or murmur would be diastolic- outside of systole (demarcated by S1 and S2).

Attachment:



Question #: 28

A 49-year-old man comes to the physician because of a 2-week history of episodic dizziness and palpitations. He says the duration varies from seconds to a few minutes. He is given a continuous monitoring device to wear for 48 hours. The post-monitoring report shows a pulse of 145/minute and blood pressure of 95/75 mm Hg, during an episode of dizziness. Using the

attached chart, which of the following changes in cardiovascular parameters is most likely during one of these episodes?

- A. A
- B. B
- C. C
- ✓D. D
- E. E
- F. F
- G. G
- H. H

Rationale: The patient's blood pressure of 95/75mmHg can be used to determine the MAP and PP. MAP is approx. 81(reduced) and Pulse pressure is 20 (reduced). The low pulse pressure is indicative of a low stroke volume. In hypotension reduced arterial pressure leads to reduced afferent baroreceptor firing rate. This leads to increased efferent sympathetic firing and reduced efferent parasympathetic stimulation. This results in vasoconstriction, tachycardia, increased contractility to increase the BP.

Attachment:

	Mean Arterial Pressure	Pulse Pressure	Stroke Volume	Baroreceptor Firing Rate
A	No change	↓	↓	↓
B	↓	No change	↓	↓
C	↓	↓	No change	↓
D	↓	↓	↓	↓
E	↑	↑	↑	↑
F	No change	↑	↑	↑
G	↑	No change	↑	↑
H	↑	↑	No change	↑

cv table.jpg

Question #: 29

A 61-year-old man comes to the physician because of a 6-month history of progressively worsening shortness of breath. Physical examination shows a late systolic crescendo murmur with a midsystolic click on auscultation of the heart. The murmur is best heard over the apex, is loudest at S1, is shortened with squatting, and is longer and more intense when venous return is decreased by standing or a Valsalva maneuver. Which of the following is the best index of the preload on this patient's heart?

- A. Mean arterial pressure

- B. Central venous pressure
- C. Pulmonary capillary wedge pressure
- D. Ventricular end-diastolic pressure
- ✓E. Ventricular end-diastolic volume

Rationale: Preload is the degree to which the

myocardium is stretched before it contracts, that is, the length of the sarcomere at the end of diastole. Recall

the Frank–Starling law that states that the energy of contraction is proportional to the initial length of the cardiac muscle fiber

(preload). In vivo, the variable most directly related to sarcomere length during end diastole is ventricular end-diastolic volume (correct choice). Blood volume can influence preload through changes in end-diastolic volume. Central venous pressure, pulmonary capillary wedge pressure, and left ventricular end-diastolic pressure are indices of preload, but end-diastolic volume is the most direct reflection of changes in end-diastolic fiber length (preload).

Both Valsalva maneuver and standing decrease venous return to the heart, thereby decreasing left ventricular diastolic filling (preload), and causing more laxity on the chordae tendineae. This allows the mitral valve to prolapse earlier in systole, leading to an earlier systolic click (i.e., closer to S1) and a longer murmur.

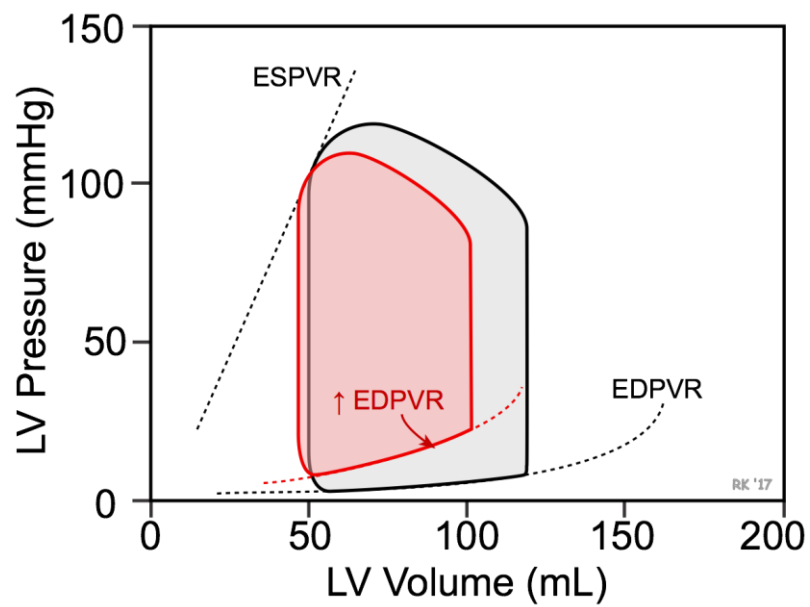
Question #: 30

A 55-year-old man is brought to the emergency department by his wife because of a 4-hour history of shortness of breath and confusion. He was diagnosed with heart failure due to diastolic dysfunction 3 years ago. Physical examination shows rales in the basal lung fields bilaterally. A pressure-volume loop (red) of the patient is shown in comparison with a normal loop (black) in the attached figure. Which of the following values best estimates ejection fraction of this patient?

- A. 20%
- B. 30%
- C. 40%
- ✓D. 50%
- E. 60%

Rationale: The correct answer is 50%. Ejection Fraction is equal to the Stroke volume (End diastolic volume – end-systolic volume) divided by the end-diastolic volume. The stroke volume of the patient is 100-50. $EF = 50/100 = 50\%$.

Attachment:



hf006-pv-loop-diastolic-dysfx.png